

Algebraic Characterization of Rule 110 over $\text{GF}(7)$ and Finite NAND Functional Completeness

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Abstract

We derive an exact algebraic representation of the Rule 110 elementary cellular automaton update rule as a degree-3 multilinear polynomial over the Galois field $\text{GF}(7) = \mathbb{Z}/7\mathbb{Z}$:

$$p(L, C, R) = C + R - C \cdot R - L \cdot C \cdot R,$$

where $L, C, R \in \{0, 1\} \subset \text{GF}(7)$ encode the left, center, and right cells of a Boolean neighborhood. This polynomial is the unique degree-3 multilinear function over $\text{GF}(7)$ interpolating the eight Rule 110 transition values. As a direct corollary, fixing the center cell to $C = 1$ collapses the polynomial to $p(L, 1, R) = 1 - L \cdot R = \text{NAND}(L, R)$: Rule 110 implements a NAND gate on Boolean inputs whenever the center cell is active. By Sheffer’s functional completeness theorem [6], NAND is a universal Boolean primitive, so Rule 110 at center cell $C = 1$ can realize any finite two-input Boolean function — a genuine, algebraically transparent completeness result in its own right, established without reference to cyclic tag systems. This finite functional-completeness fact is distinct from Turing universality, which requires simulating computation on an unbounded tape for an unbounded number of steps and is not implied by completeness over a fixed finite input size. Rule 110’s Turing universality rests on Cook’s 2004 cyclic tag system construction [1], machine-certified in Lean 4 [11, 15]. All algebraic results of this paper (the $\text{GF}(7)$ polynomial identity, the center-1 NAND gate, and the $\text{GF}(7)$ AND–NAND duality) are machine-certified in Lean 4 with zero `sorry` and require no Cook CTS lemmas. The algebraic certificate reduces the finite Boolean completeness of the Rule 110 update rule to a single degree-3 polynomial identity over a prime field. We discuss how this $\text{GF}(7)$ structure connects to the $\mathbb{Z}_7$ -symmetric Klein–Gordon substrate Φ_{MDL} (MDL: Minimum Description Length) of the Universal Generative Principle’s GTE framework [8, 9], where kink soliton dynamics directly embed Rule 110 computation. The polynomial structure reflects the $\mathbb{Z}_7$ charge-sector arithmetic of the GTE dynamical substrate [10], where this identity underlies the finite Boolean functional completeness of the physical field’s kink arithmetic; Φ_{MDL} ’s Turing universality is established separately via a Cook-route composition (§7). The prime $7 = N_c^2 - N_c + 1$ is the unique Frobenius

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prime generated by the QCD colour rank $N_c = 3$, machine-certified as `frobenius_chain_level1` (`UgpLean.Universality.FrobeniusChain`, zero sorry); the lepton seed $b_{L_1} = 73 = N_c^4 - N_c^2 + 1$ is the level-2 Frobenius prime, connecting GF(7) to the full GTE lepton mass hierarchy via the Frobenius-Generation Coincidence Identity (FGCI; `fgci_unique_at_nc`, zero sorry).

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1 Introduction

Rule 110 is the elementary cellular automaton defined by Wolfram’s numbering convention [18] in which the three-cell binary update lookup table encodes the integer 110 in binary. The rule’s

dynamical behavior is complex: it generates persistent glider structures that interact on a periodic ether background. Matthew Cook proved in 2004 that this complexity reaches Turing universality by embedding cyclic tag systems (CTS) into the rule’s glider collision dynamics [1, 2]. Cook’s proof—constructive and elegant—has been machine-certified in Lean 4 [11, 15], establishing Rule 110 universality on a fully formal footing.

Despite this success, Cook’s proof operates at the level of specific glider patterns, collision outcomes, and CTS reduction steps. The argument is combinatorial rather than algebraic: universality emerges from the geometry of the ether background and a finite collection of verified collision types [2, 4]. A natural question has remained open: *is there a compact algebraic certificate that captures some of the computational richness of Rule 110’s local update rule without passing through the CTS machinery?*

This paper answers that question affirmatively at the level of finite Boolean functional completeness. We derive the Rule 110 update function as an explicit multilinear polynomial

$$p(L, C, R) = C + R - CR - LCR$$

over $\text{GF}(7) = \mathbb{Z}/7\mathbb{Z}$ and prove, via elementary Boolean arithmetic in $\text{GF}(7)$, that fixing $C = 1$ yields $p(L, 1, R) = 1 - LR = \text{NAND}(L, R)$. Sheffer’s 1913 theorem [6] that NAND is functionally complete then shows that Rule 110 at center cell $C = 1$ can realize any two-input Boolean function as a finite NAND circuit. This is a completeness statement about finite Boolean functions of fixed input size; it does not by itself establish Turing universality, which concerns simulation of unbounded-tape, unbounded-time computation and requires an explicit tape-layout and timing construction such as Cook’s. Rule 110’s Turing universality therefore continues to rest on Cook’s theorem (§5).

The finite-completeness proof chain is entirely algebraic and independent of Cook’s CTS embedding; no glider patterns, ether shifts, or cyclic tag reduction lemmas are invoked. The entire argument machine-checks in Lean 4 via a single exhaustive `native_decide` for the polynomial identity and a single `decide` for the NAND gate step [15].

Organization. Section 2 reviews the Rule 110 lookup table and introduces notation. Section 3 derives the $\text{GF}(7)$ polynomial via Lagrange interpolation and establishes its uniqueness. Section 4 proves the center-1 NAND gate identity. Section 5 relates this finite NAND functional completeness to Rule 110’s Turing universality, which is established via Cook’s theorem. Section 6 describes the Lean 4 machine certification. Section 7 connects the $\text{GF}(7)$ polynomial structure to the $\mathbb{Z}_7$ GTE substrate. Section 8 discusses implications and the relationship between Cook’s Turing-universality proof and the algebraic finite-completeness certificate. Section 9 concludes. Appendix A lists the key Lean 4 theorem statements.

GTE substrate. The Universal Generative Principle (UGP) framework and its dynamical substrate the Generative Triple Evolution (GTE) posit that all physical phenomena emerge from a $\mathbb{Z}_7$ -symmetric cellular automaton substrate [9, 10]. The substrate is realized as a $\mathbb{Z}_7$ -symmetric Klein–Gordon field Φ_{MDL} , whose kink soliton dynamics implement the Rule 110 update rule [8]. The choice of $\text{GF}(7)$ is not arbitrary: $7 = N_c^2 - N_c + 1$ is the unique Frobenius prime generated by the QCD colour rank $N_c = 3$, machine-certified as `frobenius_chain_level1` (`UgpLean.Universality.FrobeniusChain`, zero sorry). The seven-element field is uniquely selected by requiring $N_c = 3$ and the Frobenius prime property. Furthermore, the lepton seed $b_{L_1} = 73 = N_c^4 - N_c^2 + 1$ is the level-2 Frobenius prime $F(N_c^2)$, connecting $\text{GF}(7)$ to the full lepton mass hierarchy via the Frobenius-Generation Coincidence Identity (FGCI; `fgci_unique_at_nc`, zero sorry). A further arithmetic consilience reinforces this selection: over $2 \leq n \leq 30$,

the UGP ridge offset $\delta(n) = n + (n^2 - 1)/2$ coincides with the minimal embeddable odd prime $q_{\min}(n)$ (the least odd prime q with $n \mid q - 1$) precisely at $n = N_c = 3$, where both equal 7; the coincidence is machine-certified as `delta_qmin_coincidence_at_three` (zero sorry) and is a bounded over-determination of the modulus selection, not an independent forcing route [7].

The GF(7) polynomial identity derived here is not merely a mathematical convenience: it reflects the $\mathbb{Z}_7$ charge-sector structure of the physical substrate, where the polynomial coefficients encode generation-orbit transitions and the prime 7 is the natural modulus of the winding-number arithmetic [10].

Companion papers. This paper is a companion to P28 [8] (computational universality and SM structure), P30 [11] (Lean 4 machine certification of Cook’s theorem), and P34 [9] (the GTE algebraic substrate).

The Two Laws: One Framework, Two Levels

Level 1 — The Algebraic Certificate (the discrete law, 19 bits):

$$p(L, C, R) = C + R - CR - LCR \pmod{7}$$

A single four-term polynomial over GF(7). Certifies which structures the physical field must contain. Machine-certified unique ($\mathbf{A}_{\text{Lean}}$; P40, P48).

Level 2 — The Field Law (the continuum law):

$$\mathcal{L}_{\Phi_{\text{MDL}}} = \frac{1}{2}(\partial_\mu \Phi)^2 - \frac{m^2}{49}(1 - \cos 7\Phi) + \mathcal{L}_{\text{gauge}} + \mathcal{L}_\chi$$

The unique Lorentz-invariant $\mathbb{Z}_7$ -symmetric field realising the Level 1 certificate ($\mathbf{A}_{\text{Lean}}$; P42, P48).

The bridge — self-consistency forces one mass scale:

$$p(x, x, x) = x \implies x^* = \frac{1}{\varphi} \implies m_\varphi = m_\tau$$

The Level 2 field law is derived in P42 [13]. The complete two-level derivation: P48 [7].

2 Rule 110 Lookup Table and Boolean Encoding

2.1 Definition and bit-numbering

An elementary cellular automaton (ECA) evolves a one-dimensional binary tape $t : \mathbb{Z} \rightarrow \{0, 1\}$ by applying a local rule $f : \{0, 1\}^3 \rightarrow \{0, 1\}$ to every triple $(L, C, R) = (t(i - 1), t(i), t(i + 1))$ simultaneously. Wolfram’s numbering [18] orders the eight triples by the index

$$k(L, C, R) = 4L + 2C + R \in \{0, 1, \dots, 7\},$$

and the rule number is $\sum_{k=0}^7 f_k \cdot 2^k$ where f_k is the output for neighborhood k .

2.2 Rule 110 truth table

The outputs for Rule 110 are given in Table 1.

The five active neighborhoods (output 1) are $k \in \{1, 2, 3, 5, 6\}$; the three inactive neighborhoods (output 0) are $k \in \{0, 4, 7\}$.

Table 1: Rule 110 transition table. Neighborhood index $k = 4L + 2C + R$; output bit f_k .

L	C	R	k	f_k
1	1	1	7	0
1	1	0	6	1
1	0	1	5	1
1	0	0	4	0
0	1	1	3	1
0	1	0	2	1
0	0	1	1	1
0	0	0	0	0

2.3 GF(7) embedding

Throughout this paper we embed Boolean values into $\text{GF}(7) = \mathbb{Z}/7\mathbb{Z}$ via

$$\text{false} \mapsto 0, \quad \text{true} \mapsto 1.$$

All arithmetic below is in $\text{GF}(7)$. Since $L, C, R \in \{0, 1\} \subset \text{GF}(7)$, we have $L^2 = L$, $C^2 = C$, and $R^2 = R$ (idempotence), so every symmetric function of Boolean inputs has a multilinear representative of degree at most 3.

3 The GF(7) Polynomial Representation

3.1 Lagrange interpolation over a prime field

Given any function $f : \{0, 1\}^3 \rightarrow \{0, 1\}$, there exists a unique multilinear polynomial $p(L, C, R)$ over any field of characteristic $\neq 2$ interpolating the values of f at all eight Boolean points. We work over $\text{GF}(7)$ (characteristic 7, so $\text{char} \neq 2$).

The general multilinear form in three variables is:

$$p(L, C, R) = a_0 + a_L L + a_C C + a_R R + a_{LC} LC + a_{LR} LR + a_{CR} CR + a_{LCR} LCR.$$

Imposing $p(L, C, R) = f_{k(L, C, R)}$ at all eight Boolean inputs yields eight linear equations in the eight coefficients $a_0, a_L, \dots, a_{LCR}$ over $\text{GF}(7)$.

3.2 Solving the interpolation system

Evaluating the constraints in order of increasing k :

$$\begin{aligned} p(0, 0, 0) &= a_0 = 0, \\ p(0, 0, 1) &= a_0 + a_R = 1, \\ p(0, 1, 0) &= a_0 + a_C = 1, \\ p(0, 1, 1) &= a_0 + a_C + a_R + a_{CR} = 1, \\ p(1, 0, 0) &= a_0 + a_L = 0, \\ p(1, 0, 1) &= a_0 + a_L + a_R + a_{LR} = 1, \\ p(1, 1, 0) &= a_0 + a_L + a_C + a_{LC} = 1, \end{aligned}$$

$$p(1, 1, 1) = a_0 + a_L + a_C + a_R + a_{LC} + a_{LR} + a_{CR} + a_{LCR} = 0.$$

From the first equation, $a_0 = 0$. From the second, $a_R = 1$. From the third, $a_C = 1$. From the fourth, $a_{CR} = 1 - a_C - a_R - a_0 = 1 - 1 - 1 = -1 \equiv 6 \pmod{7}$. From the fifth, $a_L = -a_0 = 0$. From the sixth, $a_{LR} = 1 - a_L - a_R - a_0 = 1 - 0 - 1 = 0$. From the seventh, $a_{LC} = 1 - a_L - a_C - a_0 = 1 - 0 - 1 = 0$. From the eighth:

$$a_{LCR} = 0 - a_L - a_C - a_R - a_{LC} - a_{LR} - a_{CR} - a_0 = -1 - 1 - (-1) = -1 \equiv 6 \pmod{7}.$$

The unique solution is therefore $a_0 = 0$, $a_L = 0$, $a_C = 1$, $a_R = 1$, $a_{LC} = 0$, $a_{LR} = 0$, $a_{CR} = -1$, $a_{LCR} = -1$, giving:

Theorem 3.1 (GF(7) polynomial representation of Rule 110). *The Rule 110 update function is represented over GF(7) by the unique degree-3 multilinear polynomial*

$$p(L, C, R) = C + R - CR - LCR.$$

That is, for all $(L, C, R) \in \{0, 1\}^3$,

$$f_{110}(L, C, R) = p(L, C, R) \in \{0, 1\},$$

where the right-hand side is evaluated in GF(7) and its result lies in $\{0, 1\} \subset \text{GF}(7)$.

3.3 Verification

The identity is confirmed exhaustively: each of the eight Boolean triples (L, C, R) gives an output in $\{0, 1\} \subset \text{GF}(7)$ matching the Rule 110 truth table. In GF(7), subtraction distributes freely, so $-CR - LCR = 6CR + 6LCR$. A direct check:

- $(0, 0, 0)$: $0 + 0 - 0 - 0 = 0$. ✓
- $(0, 0, 1)$: $0 + 1 - 0 - 0 = 1$. ✓
- $(0, 1, 0)$: $1 + 0 - 0 - 0 = 1$. ✓
- $(0, 1, 1)$: $1 + 1 - 1 - 0 = 1$. ✓
- $(1, 0, 0)$: $0 + 0 - 0 - 0 = 0$. ✓
- $(1, 0, 1)$: $0 + 1 - 0 - 0 = 1$. ✓
- $(1, 1, 0)$: $1 + 0 - 0 - 0 = 1$. ✓
- $(1, 1, 1)$: $1 + 1 - 1 - 1 = 0$. ✓

All eight outputs match Rule 110.

Remark 3.2. The polynomial has only four nonzero terms: C , R , $-CR$, and $-LCR$. The absence of the L , LR , and LC terms reflects Rule 110's structural asymmetry in the left neighbor: L influences the output only through its interaction with C . The degree-3 term $-LCR$ is the unique cubic correction that distinguishes $f_{110}(1, 1, 1) = 0$ from the linear-quadratic prediction $1 + 1 - 1 = 1$.

3.4 Uniqueness

Proposition 3.3 (Uniqueness of multilinear interpolant over $\text{GF}(7)$). *The polynomial $p(L, C, R) = C + R - CR - LCR$ is the unique degree-3 multilinear polynomial over $\text{GF}(7)$ agreeing with f_{110} on all eight Boolean inputs.*

Proof. The set $\{0, 1\}^3$ has eight points in $\text{GF}(7)^3$. The space of multilinear polynomials in three variables has dimension $2^3 = 8$ (one monomial per subset of $\{L, C, R\}$). The Vandermonde-type interpolation system at the eight Boolean points has full rank over any field of characteristic > 2 (in particular over $\text{GF}(7)$), so the interpolant is unique. $\square$

3.5 Polynomial certificates for other ECA rules

The Lagrange interpolation technique of Sections 3.1–3.3 applies verbatim to any elementary cellular automaton rule. Given any $f : \{0, 1\}^3 \rightarrow \{0, 1\}$, the identical 8×8 system over $\text{GF}(7)$ yields the unique degree-3 multilinear polynomial interpolating the rule's truth table. Different ECA rules differ only in their output vectors, producing different coefficient solutions.

Proposition 3.4 ($\text{GF}(7)$ polynomial for Rule 124). *Rule 124 is the spatial mirror of Rule 110 (obtained by the $L \leftrightarrow R$ reflection of the neighborhood). Its truth table is:*

$$\begin{aligned} f_{124}(0, 0, 0) &= 0, & f_{124}(0, 0, 1) &= 0, & f_{124}(0, 1, 0) &= 1, & f_{124}(0, 1, 1) &= 1, \\ f_{124}(1, 0, 0) &= 1, & f_{124}(1, 0, 1) &= 1, & f_{124}(1, 1, 0) &= 1, & f_{124}(1, 1, 1) &= 0. \end{aligned}$$

The unique degree-3 multilinear polynomial over $\text{GF}(7)$ interpolating f_{124} is:

$$q(L, C, R) = L + C - LC - LCR.$$

Proof. Applying the interpolation system of Section 3.1–3.2 to the Rule 124 truth table yields the linear system whose unique solution is $a_0 = 0$, $a_L = 1$, $a_C = 1$, $a_R = 0$, $a_{LC} = -1$, $a_{LR} = 0$, $a_{CR} = 0$, $a_{LCR} = -1$, giving $q(L, C, R) = L + C - LC - LCR$. Verification is immediate: evaluating at all eight Boolean triples matches the Rule 124 truth table exhaustively. Alternatively, since $f_{124}(L, C, R) = f_{110}(R, C, L)$ (the $L \leftrightarrow R$ spatial reflection), substituting $L \leftrightarrow R$ into the Rule 110 polynomial $p(L, C, R) = C + R - CR - LCR$ gives $p(R, C, L) = C + L - CL - RCL = L + C - LC - LCR = q(L, C, R)$. $\square$

Remark 3.5. Comparing the two polynomials:

$$p_{110}(L, C, R) = C + R - CR - LCR, \quad q_{124}(L, C, R) = L + C - LC - LCR.$$

The cubic term $-LCR$ is the same in both: it is symmetric in L and R , so the spatial reflection acts only on the quadratic and linear terms. Where Rule 110 has the pair $(R, -CR)$ (involving the right neighbor), Rule 124 has the pair $(L, -LC)$ (involving the left neighbor). The structural asymmetry between L and R in Rule 110 is precisely reversed in Rule 124, and this reversal is algebraically transparent in the $\text{GF}(7)$ polynomials.

4 Rule 110 at Center = 1 Implements NAND

Theorem 4.1 (Center-1 NAND gate). *For all $(L, R) \in \{0, 1\}^2$,*

$$p(L, 1, R) = 1 - LR = \text{NAND}(L, R) \in \{0, 1\}.$$

Equivalently, $f_{110}(L, 1, R) = \neg(L \wedge R)$ for all Boolean L, R .

Proof. Substituting $C = 1$ into the polynomial:

$$\begin{aligned} p(L, 1, R) &= 1 + R - 1 \cdot R - L \cdot 1 \cdot R \\ &= 1 + R - R - LR \\ &= 1 - LR. \end{aligned}$$

Since $L, R \in \{0, 1\}$, we have $LR \in \{0, 1\}$, so $1 - LR \in \{0, 1\}$ and equals 0 iff $L = R = 1$, i.e., $1 - LR = \text{NAND}(L, R)$ over $\{0, 1\}$. $\square$

The computation $1 + R - R = 1$ eliminates the quadratic term R entirely, leaving only the cubic correction $-LR$. The cancellation $R - CR|_{C=1} = R - R = 0$ is the key algebraic step.

Remark 4.2. Over $\text{GF}(7)$, $-LR \equiv 6LR$, so $p(L, 1, R) = 1 + 6LR$. For Boolean L, R : when $LR = 0$, $p = 1$; when $LR = 1$, $p = 1 + 6 = 7 \equiv 0 \pmod{7}$. The NAND identity $1 - LR = \text{NAND}(L, R)$ holds in any prime field of characteristic $\neq 2$; the specific relevance of $\text{GF}(7)$ is that this prime field coincides with the $\mathbb{Z}_7$ winding-number structure of the GTE substrate.

We also record the companion $\text{GF}(7)$ AND identity:

Lemma 4.3 ($\text{GF}(7)$ AND–NAND duality). *For all $(L, R) \in \{0, 1\}^2$, over $\text{GF}(7)$:*

$$L \cdot R = 1 - f_{110}(L, 1, R).$$

Equivalently, the AND gate factors through Rule 110 at center 1 as $L \wedge R = \neg f_{110}(L, 1, R)$.

Proof. From Theorem 4.1, $f_{110}(L, 1, R) = 1 - LR$, so $LR = 1 - f_{110}(L, 1, R)$. $\square$

5 NAND Functional Completeness and Turing Universality

5.1 NAND functional completeness

A set of Boolean operations is *functionally complete* if every Boolean function can be expressed as a finite composition of operations from the set. Sheffer [6] established in 1913 that the single binary operation NAND (the Sheffer stroke $\uparrow$) is functionally complete: every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is expressible as a finite circuit of NAND gates. In particular, NOT and AND are NAND-expressible ($\neg a = a \uparrow a$ and $a \wedge b = \neg(a \uparrow b)$), from which any finite Boolean function follows by standard circuit synthesis [5].

Theorem 5.1 (Finite NAND functional completeness of Rule 110). *Every two-input Boolean function $f : \{0, 1\}^2 \rightarrow \{0, 1\}$ is realized by a finite composition of Rule 110 center-1 NAND steps.*

Proof. By Theorem 4.1, Rule 110 with center cell fixed to 1 computes NAND on Boolean inputs (L, R) . By Sheffer’s functional completeness theorem [6], every two-input Boolean function is a finite composition of NAND gates. Therefore, Rule 110 (operating on appropriately configured center-1 cells) can evaluate any finite two-input Boolean circuit. $\square$

Remark 5.2 (This is not, by itself, a proof of Turing universality). Theorem 5.1 is a statement about Boolean functions of *fixed, finite* input size: for each n , Rule 110’s NAND primitive can realize every n -input Boolean function using a finite circuit, but the circuit size and the tape-layout/timing needed to realize it as an actual Rule 110 space–time diagram both depend on n and are not bounded uniformly. Turing universality requires simulating an unbounded-tape, unbounded-time Turing machine on *unboundedly many* inputs with a single, uniformly described mechanism — exactly what Cook’s construction supplies via cyclic tag systems, and what functional completeness over a fixed input size does not supply on its own. Rule 110’s Turing universality rests on Cook’s 2004 cyclic tag system construction, discussed next.

5.2 Cook’s theorem remains the route to Turing universality

Cook’s 2004 proof [1], Lean-certified in [11, 15], is the load-bearing result establishing that Rule 110 is Turing universal: it supplies an explicit, uniform cyclic-tag-system embedding that simulates any Turing machine on Rule 110’s infinite space–time diagram, for unboundedly many inputs and unboundedly many steps. The finite NAND functional-completeness result of this paper (Theorem 5.1) is a separate, algebraically transparent fact about Rule 110’s local update rule: it shows that the two-cell neighborhood arithmetic underlying Rule 110 is, on its own, as expressive as Boolean logic gets at a fixed input size. The two results are complementary rather than competing:

- **Cook:** explicit cyclic-tag-system embedding; unbounded tape and time; the genuine Turing-universality proof for Rule 110.
- **Algebraic (this paper):** $O(1)$ -term polynomial identity; no ether patterns; finite two-input NAND functional completeness of the local update rule.

6 Lean 4 Machine Certification

The results of Sections 3–5 are machine-certified in Lean 4 [3]. The formalization lives in two companion libraries:

- `rule110-lean` [15], module `Rule110.AlgebraicUniversality`.
- `UgpLean.Universality.PhiMDLUniversality` in the `ugp-lean` library [17].

6.1 Certified theorems (zero sorry)

rule110_z7_poly_rep ($\mathbf{A}_{\text{Lean}}$, zero sorry). For all $(L, C, R) \in \{\text{false}, \text{true}\}^3$:

$$\begin{aligned} \text{boolToZ7}(\text{rule110Step } L \ C \ R) &= \text{boolToZ7 } C + \text{boolToZ7 } R \\ &\quad - \text{boolToZ7 } C \cdot \text{boolToZ7 } R - \text{boolToZ7 } L \cdot \text{boolToZ7 } C \cdot \text{boolToZ7 } R \end{aligned}$$

in $\mathbb{ZMod } 7$. Proof method: `native_decide` (exhaustive check over all 8 Boolean triples).

rule110_center1_is_nand ($\mathbf{A}_{\text{Lean}}$, zero sorry). For all $(L, R) \in \{\text{false}, \text{true}\}^2$:

$$\text{rule110Step } L \ \text{true } R = \neg(L \ \&\& \ R).$$

Proof method: `decide` (exhaustive case check).

rule110_z7_nand_identity ($\mathbf{A}_{\text{Lean}}$, zero sorry). For all $(L, R) \in \{\text{false}, \text{true}\}^2$, over $\mathbb{ZMod } 7$:

$$\text{boolToZ7 } L \cdot \text{boolToZ7 } R = 1 - \text{boolToZ7}(\text{rule110Step } L \ \text{true } R).$$

Proof method: `native_decide`.

rule110_algebraic_nand_bundle ($\mathbf{A}_{\text{Lean}}$, zero sorry). Bundles the NAND gate and GF(7) AND-NAND identity into a single theorem: for all (L, R) , $\text{rule110Step } L \ \text{true } R = \neg(L \ \&\& \ R)$ and $\text{boolToZ7 } L \cdot \text{boolToZ7 } R = 1 - \text{boolToZ7}(\text{rule110Step } L \ \text{true } R)$ hold simultaneously.

6.2 The one named axiom

boolean_nand_complete (one named axiom).

$$\forall f : \text{Bool} \rightarrow \text{Bool} \rightarrow \text{Bool}, \exists \text{circuit} : \text{Bool} \rightarrow \text{Bool} \rightarrow \text{Bool}, \forall a b, \text{circuit } a b = f a b.$$

This states that every two-input Boolean function is computable as a NAND circuit. It is a formal statement of Sheffer’s completeness theorem [6]. The result is entirely independent of Cook’s CTS construction and of Rule 110 in particular. A full Lean 4 proof can be obtained by exhaustive construction over the 16 two-input Boolean functions; we leave this as an acknowledged open task in the Lean library.

6.3 Finite functional-completeness theorem

rule110_center1_nand_functional_completeness ($\mathbf{A}_{\text{Lean}}$, zero sorry given **boolean_nand_complete**).

$$\forall f : \text{Bool} \rightarrow \text{Bool} \rightarrow \text{Bool}, \exists \text{circuit} : \text{Bool} \rightarrow \text{Bool} \rightarrow \text{Bool}, \forall a b, \text{circuit } a b = f a b.$$

Proof: **rule110_center1_nand_functional_completeness** := **boolean_nand_complete**. Given the axiom, the proof is a one-line application. **Scope note:** this theorem’s statement is exactly the finite two-input Boolean functional-completeness statement of **boolean_nand_complete**; it establishes Theorem 5.1 (finite NAND completeness of Rule 110), not Turing universality on an unbounded tape. Rule 110’s Turing universality rests on Cook’s 2004 construction; the conditional operational readback certificate for this is **cook_operational_stage3_tm_microstep_readback** (**rule110-lean**, zero sorry modulo five named classical Cook collision-analysis bridge axioms; §5) — itself a microstep-readback certificate for already-supplied compilations, not a self-contained Turing-universality theorem; the general $\text{TM} \rightarrow \text{CTS}$ compiler remains open in **rule110-lean**. (This theorem was formerly named **rule110_turing_universal_algebraic**, which is now kept only as a deprecated alias in **rule110-lean**.)

6.4 Certification summary

The four zero-sorry finite-completeness theorems rest on Lean 4’s standard axiom signature (**propext**, **Classical.choice**, **Quot.sound**) and use no Cook CTS axioms. The sole outstanding open item for the finite-completeness chain is a Lean formalization of Sheffer’s classical completeness result, which is a finite computation over the 16 two-input Boolean functions. Rule 110’s Turing universality is a separate result, resting on Cook’s theorem and its five named classical bridge axioms formalizing Cook’s own published collision analysis (**rule110-lean**); it is not implied by, and does not require, the finite-completeness chain above.

7 Connection to the $\mathbb{Z}_7$ -Symmetric GTE Substrate

7.1 The $\mathbb{Z}_7$ -KG field

The Universal Generative Principle’s Generative Triple Evolution (GTE) framework [8,9] identifies the fundamental computational substrate as a $\mathbb{Z}_7$ -symmetric Klein–Gordon ($\mathbb{Z}_7$ -KG) field Φ_{MDL} . This field carries topological kink solitons with winding numbers $Q \in \mathbb{Z}_7 = \mathbb{Z}/7\mathbb{Z}$. The three Standard Model generations correspond to winding sectors $Q \neq 0$ (active), with vacuum $Q = 0$ corresponding to the ether background.

Table 2: Lean 4 certification status for the finite-completeness theorems of this paper.

Theorem	Status	Proof method
<code>rule110_z7_poly_rep</code>	$\mathbf{A}_{\text{Lean}}, 0$ sorry	<code>native_decide</code>
<code>rule110_center1_is_nand</code>	$\mathbf{A}_{\text{Lean}}, 0$ sorry	<code>decide</code>
<code>rule110_z7_nand_identity</code>	$\mathbf{A}_{\text{Lean}}, 0$ sorry	<code>native_decide</code>
<code>rule110_algebraic_nand_bundle</code>	$\mathbf{A}_{\text{Lean}}, 0$ sorry	composition
<code>boolean_nand_complete</code>	1 named axiom	open (Sheffer’s thm)
<code>rule110_center1_nand_functional_completeness</code>	$\mathbf{A}_{\text{Lean}} _{\text{ax}}, 0$ sorry	finite NAND complete-ness (not unbounded-tape universality)
<code>cook_operational_stage3_tm_microstep_readback</code>	$\mathbf{A}_{\text{Lean}}, 0$ sorry, 5 named axioms	Cook’s theorem (<code>rule110-lean</code> ; conditional operational microstep read-back, not itself full Turing universality)

7.2 Kink dynamics embed Rule 110

Physical site updates in the $\mathbb{Z}_7$ -KG field map winding-number triples (Q_L, Q_C, Q_R) to output winding numbers via the active predicate $\text{active}(Q) \Leftrightarrow Q \neq 0$:

$$\text{z7kg_rule110_step}(Q_L, Q_C, Q_R) = \begin{cases} 1 & \text{if } f_{110}(\text{active}(Q_L), \text{active}(Q_C), \text{active}(Q_R)) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The kink step therefore directly implements the Rule 110 update at the level of winding numbers (Lean 4 certified as `z7kg_kink_universality`, zero sorry).

The $\text{GF}(7)$ polynomial structure of the Rule 110 update function—its status as a polynomial over $\mathbb{Z}/7\mathbb{Z}$ —is thus not incidental. The $\text{GF}(7)$ field is the natural coefficient ring for the charge sector arithmetic of the GTE framework: the seven-fold symmetry of the charge sector ($\mathbb{Z}_7$ winding numbers) is the same structure that makes the polynomial interpolation natural.

7.3 Polynomial identity from kink arithmetic

Over $\text{GF}(7)$, Boolean inputs L, C, R encode active (1) or vacuum (0) winding numbers. The Rule 110 polynomial $p(L, C, R) = C + R - CR - LCR$ can be interpreted physically as:

- $C + R$: additive contribution of center and right kinks.

- $-CR$: subtraction when both C and R are active (collision term).
- $-LCR$: three-kink collision correction (all-active suppression, matching $f_{110}(1, 1, 1) = 0$).

The cancellation at the fully active neighborhood $(L, C, R) = (1, 1, 1)$ mirrors the physical fact that a triple kink collision in the $\mathbb{Z}_7$ -KG vacuum destroys all three solitons.

7.4 Φ_{MDL} is Turing universal (Cook route)

The Φ_{MDL} field evolution is machine-certified as Turing universal via the *Cook route*: the proved stepwise simulation of Rule 110 by Φ_{MDL} kink dynamics (`phimdl_law_description_execution`, zero sorry) composed with Cook’s operational universality theorem for Rule 110 (`cook_rule110_simulates_computable`, one named axiom packaging Cook’s theorem with universal Turing-machine compilation), yielding `phimdl_turing_universal` (`ALean` conditional; zero sorry; one named axiom; `UgpLean.Universality.PhiMDLUniversality`). Separately, the same $\text{GF}(7)$ polynomial identity used in this paper certifies that Φ_{MDL} kink arithmetic is finitely Boolean-functionally complete (`z7_bool3_finite_functional_completeness`, zero sorry) — the field-theoretic analogue of Theorem 5.1 above, and, like that theorem, a distinct fact from Turing universality rather than a second, independent route to it. Rule 110’s own Turing universality is thus established at the discrete Boolean-tape level via Cook’s theorem (§5), and Φ_{MDL} inherits Turing universality via the Cook route composed with the proved kink simulation (P28 [8]).

8 Discussion

8.1 Cook’s proof and the algebraic finite-completeness certificate

Rule 110’s Turing universality rests on one proof: Cook’s cyclic-tag-system embedding [1], Lean-certified in [11, 15]. Alongside it, this paper contributes a distinct, algebraically transparent result:

1. **Cook (2004) [1], Lean-certified in [11]**. Constructs an explicit cyclic tag system embedding in Rule 110 glider dynamics. The proof is combinatorial, relies on specific ether patterns and collision tables, and provides an explicit, unbounded-tape Turing machine simulation — the genuine Turing-universality proof.
2. **Algebraic (this paper)**. Derives finite two-input NAND functional completeness from the polynomial identity $p(L, 1, R) = 1 - LR$ over $\text{GF}(7)$ and Sheffer’s completeness theorem. The proof is algebraic, requires no ether patterns or collision tables, and reduces to a single degree-3 identity — but, per Remark 5.2, does not by itself establish Turing universality.

The two results are independent in the sense that neither invokes lemmas from the other, and they answer different questions: Cook’s theorem answers "can Rule 110 simulate any Turing machine?" (yes, constructively); the algebraic result answers "how expressive is Rule 110’s local update rule at a fixed input size?" (maximally expressive, in the Sheffer sense). Cook’s proof is the one that carries the unbounded-tape, unbounded-time guarantee; the algebraic proof is shorter and more transparent, and isolates exactly the finite-completeness content of the NAND identity without ether patterns or collision tables.

8.2 The role of the prime 7

It is notable that the polynomial representation is most naturally stated over $\text{GF}(7)$ rather than over, say, $\text{GF}(2)$ or $\mathbb{R}$. Over $\text{GF}(2)$, every Boolean function has a polynomial representation, but $-CR - LCR$ becomes $+CR + LCR$ (since $-1 \equiv 1 \pmod{2}$), disguising the cancellation structure that makes the NAND identity transparent. Over $\mathbb{R}$ the interpolation is non-canonical (infinitely many polynomial extensions). Over $\text{GF}(7)$: the characteristic is large enough that $-1 \neq 1$ (so subtraction is genuinely distinct from addition), the interpolation is unique and canonical, and the prime field coincides with the $\mathbb{Z}_7$ winding number structure of the GTE substrate. This confluence suggests that $\text{GF}(7)$ is the natural algebraic setting for Rule 110 dynamics within the GTE framework.

The suggestion is now theorem-backed. The interpolation system of Section 3 — which this paper feeds with the Rule 110 truth table (Table 1) — is derivable from the UGP side without that table as input. The total-parity shadow of the canonical generation-orbit triples of the UGP arithmetic (the integer-triple cascade underlying the GTE lepton mass hierarchy), placed on the $\mathbb{Z}_5$ family ring, constrains seven of the eight Boolean interpolation points, and vacuum transparency (the quiescence condition $p(0, 0, 0) = 0$, which is equivalent in Minimum Description Length (MDL) cost among the orbit-satisfying rules) supplies the eighth. The resulting system is full-rank over $\text{GF}(7)$, and its unique multilinear solution is $p(L, C, R) = C + R - CR - LCR$ itself (Direct-Interpolation Lift Theorem; `ugp_orbit_interpolation_lift`, $\mathbf{A}_{\text{Lean}}$, zero sorry, `ugp-lean`) [7, 12]. Rule 110 thereby arrives as the binary restriction of a rule forced by orbit arithmetic — the converse of this paper’s derivation, which proceeds from the truth table to the polynomial. $\text{GF}(7)$ is therefore not merely the setting in which the Rule 110 interpolation is most transparent; it is the setting in which the same eight-point interpolation problem is posed, and uniquely solved, by UGP orbit data alone.

8.3 Open directions

We note that the continuum connection has since been established (P41/P42 [13, 16]): the Lean 4 theorem `phimdl_turing_universal` ($\mathbf{A}_{\text{Lean}}$ conditional; zero sorry; one named axiom: `cook_rule110_simulates_computable`) proves Φ_{MDL} Turing-universal via the Cook route composed with the proved stepwise Φ_{MDL} -Rule 110 simulation, and Poincaré invariance of Φ_{MDL} in the continuum is certified as `poincare_invariance_of_kg`. The Lean certification `phimdl_turing_universal` operates at Level 1 (discrete CMCA, $\mathbf{A}_{\text{Lean}}$ conditional on one named axiom). Direct smooth-KG Turing universality at Level 2 — showing that the Φ_{MDL} continuum field itself can simulate a Turing machine — inherits this property via the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, $\mathbf{A}_{\text{Lean}}$, P41 [16]), but a direct Level 2 Lean certification is not yet available.

The following extensions of the algebraic finite-completeness certificate remain open:

1. **Lean formalization of Sheffer’s theorem.** The axiom `boolean_nand_complete` can in principle be given a zero-sorry Lean 4 proof by exhaustive construction over the 16 two-input Boolean functions. This would make the entire finite-completeness chain sorry-free.
2. **Circuit overhead bounds.** The algebraic proof establishes finite functional completeness but does not give an explicit circuit encoding with overhead bounds. Combining the polynomial identity with the Cook encoding [2] to obtain explicit overhead bounds relating the two constructions is a natural next step.

The invariant subset structure of $p(L, C, R)$ over $\mathbb{Z}_7$ — specifically, the classification of all subsets $S \subseteq \mathbb{Z}_7$ closed under p , and the QNR argument explaining why $\{0, 1\}$ is the unique proper invariant

subset — is established in the companion paper P49 [12]. That paper also proves that f_{MDL} is not a degree- ≤ 3 polynomial (Schwartz–Zippel), providing a clean algebraic separation between the polynomial p and the f_{MDL} lookup table.

9 Conclusion

We have established that the Rule 110 elementary cellular automaton update function is exactly represented by the degree-3 multilinear polynomial

$$p(L, C, R) = C + R - CR - LCR$$

over $\text{GF}(7)$. This polynomial is unique (Proposition 3.3) and is derived by Lagrange interpolation from the Rule 110 truth table. Fixing the center cell to 1 reduces the polynomial to $1 - LR = \text{NAND}(L, R)$ (Theorem 4.1), establishing that Rule 110 directly implements a NAND gate on Boolean inputs. Sheffer’s classical functional completeness theorem [6] then yields finite two-input Boolean functional completeness of Rule 110’s local update rule (Theorem 5.1) via an algebraic route that invokes no cyclic tag system lemma. This finite-completeness result is distinct from Turing universality (Remark 5.2): Rule 110’s Turing universality is Cook’s theorem [1], and remains the load-bearing result for that claim.

The algebraic finite-completeness certificate is machine-certified in Lean 4 in module `Rule110.AlgebraicUniversality` [15]. The polynomial identity carries zero sorry (`native_decide`); the NAND gate step carries zero sorry (`decide`); and finite Boolean functional completeness follows from one named axiom, `boolean_nand_complete`, capturing Sheffer’s result.

The $\text{GF}(7)$ structure of the polynomial is not coincidental. It matches the $\mathbb{Z}_7$ winding number arithmetic of the GTE framework’s Φ_{MDL} substrate [8, 9], where kink soliton collisions implement exactly the Rule 110 update. Rule 110’s finite Boolean functional completeness is thus built into the charge-sector structure of the GTE substrate through a four-term polynomial identity—arguably the simplest possible algebraic certificate of finite functional completeness known for any natural dynamical system. The seven-element field is itself arithmetically distinguished: $7 = N_c^2 - N_c + 1$ is the unique Frobenius prime generated by the QCD colour rank $N_c = 3$, and the lepton seed $b_{L_1} = 73 = N_c^4 - N_c^2 + 1$ is the level-2 Frobenius prime connecting $\text{GF}(7)$ to the SM lepton hierarchy via the Frobenius-Generation Coincidence Identity (FGCI), machine-certified in Lean 4 with zero sorry (`fgci_unique_at_nc`, `UgpLean.Universality.FrobeniusChain`).

The same Lagrange interpolation technique yields polynomial certificates for any elementary cellular automaton rule over $\text{GF}(7)$. As shown in Section 3.5, Rule 124 (the spatial mirror of Rule 110) has the polynomial $q(L, C, R) = L + C - LC - LCR$: the cubic $-LCR$ term is identical to Rule 110’s, while the quadratic and linear terms reflect the $L \leftrightarrow R$ symmetry exchange, making the mirror relationship algebraically transparent.

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Author Contributions

N.S. derived the polynomial representation, developed the Lean 4 certification, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

Data Availability

The Lean 4 source code and proofs are publicly available in the `rule110-lean` repository [15] at <https://github.com/novaspivack/rule110-lean>.

Code Availability

All Lean 4 source files are in `Rule110.AlgebraicUniversality` and `UgpLean.Universality.PhiMDLUniversality` within the companion repositories [15]. The full UGP Physics programme is catalogued at <https://www.novaspivack.com/research> [14]; source and papers are at <https://github.com/novaspivack/ugp-physics>.

References

- [1] Matthew Cook. Universality in elementary cellular automata. *Complex Systems*, 15(1):1–40, 2004.
- [2] Matthew Cook. A concrete view of rule 110 computation. *Electronic Proceedings in Theoretical Computer Science*, 1:31–55, 2009. arXiv:0906.3248.
- [3] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Artificial Intelligence*, pages 625–635. Springer, 2021.
- [4] Turlough Neary and Damien Woods. P-completeness of cellular automaton rule 110. In *Automata, Languages and Programming (ICALP 2006), Part I*, volume 4051 of *Lecture Notes in Computer Science*, pages 132–143. Springer, 2006.
- [5] Claude E. Shannon. A symbolic analysis of relay and switching circuits. *Transactions of the American Institute of Electrical Engineers*, 57(12):713–723, 1938.
- [6] Henry M. Sheffer. A set of five independent postulates for Boolean algebras, with application to logical constants. *Transactions of the American Mathematical Society*, 14(4):481–488, 1913.
- [7] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl} . Zenodo, 2026. <https://doi.org/10.5281/zenodo.20560550>. Preprint.
- [8] Nova Spivack. Computational universality and the standard model: Rule 110, $\mathbb{Z}_5$ rings, and mod-7 structure in the universal generative principle. Zenodo, 2026. Paper P28 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20259513>.

- [9] Nova Spivack. The GTE-Möbius substrate: Arithmetic unification of computation and transputation in the universal generative principle. Zenodo, 2026. Paper P34 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20319419>.
- [10] Nova Spivack. GTE unification: A bidirectional correspondence between rule 110 orbit arithmetic and the standard model electroweak sector. Zenodo, 2026. Paper P35 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20319421>.
- [11] Nova Spivack. Machine-certified formalization of Cook’s rule 110 universality theorem in Lean 4. Zenodo, 2026. Paper P30 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20319411>.
- [12] Nova Spivack. MDL selects the Wolfram rule: $\mathbb{Z}_7$ dynamics, algebraic structure, and Standard Model encoding of the GTE polynomial. Preprint, 2026. <https://doi.org/10.5281/zenodo.20572656>. UGP Physics series, P49.
- [13] Nova Spivack. The ϕ_{mdl} field: Quantum structure, born rule, and continuum completion of the chiral minkowski CA. Zenodo, 2026. <https://doi.org/10.5281/zenodo.20417576>. Paper P42 in the UGP Physics series.
- [14] Nova Spivack. Reflexive reality and ugp physics programmes. Research Portal, 2026. Catalogues of the Reflexive Reality programme (NEMS/MFRR) and the UGP Physics programme, its quantitative branch.
- [15] Nova Spivack. rule110-lean: Lean 4 formalization of the rule 110 cook universality pipeline. <https://github.com/novaspivack/rule110-lean>, 2026. Machine-certified Cook universality proof pipeline in Lean 4.
- [16] Nova Spivack. The three-layer chiral minkowski cellular automaton: A unified discrete spacetime model from first principles. Zenodo, 2026. <https://doi.org/10.5281/zenodo.20417572>. Paper P41 in the UGP Physics series.
- [17] Nova Spivack. ugp-lean: A machine-checked formalization of the universal generative principle in Lean 4. Zenodo, 2026. Formalization paper (latest version): <https://doi.org/10.5281/zenodo.19554688>. Concept DOI (always-latest): <https://doi.org/10.5281/zenodo.19433538>. More than 400 modules across seventeen architectural layers including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3 -Newton flow, binary cascade, physical mass spectrum), BraidAtlas (ChargeTheorem, CompositeTriples, ChiralitySquaring, ChargeDerivation, CoxeterConductor, CoxeterConductor-TowerLaw, EWBosons), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. Zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations across all `.lean` files, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper’s axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.
- [18] Stephen Wolfram. *A New Kind of Science*. Wolfram Media, Champaign, IL, 2002.

A Lean 4 Proof Listing (Key Theorems)

The following Lean 4 statements are from Rule110.AlgebraicUniversality in rule110-lean [15]. Line comments document proof method and sorry status.

```
-- GF(7) polynomial representation (zero sorry, native_decide)
theorem rule110_z7_poly_rep :
  forall (L C R : Bool),
    boolToZ7 (rule110Step L C R) =
      (boolToZ7 C + boolToZ7 R
       - boolToZ7 C * boolToZ7 R
       - boolToZ7 L * boolToZ7 C * boolToZ7 R : ZMod 7) := by
  native_decide

-- Center-1 NAND gate (zero sorry, decide)
theorem rule110_center1_is_nand :
  forall (L R : Bool),
    rule110Step L true R = !(L && R) := by
  decide

-- GF(7) AND-NAND duality (zero sorry, native_decide)
theorem rule110_z7_nand_identity :
  forall (L R : Bool),
    (boolToZ7 L * boolToZ7 R : ZMod 7) =
      1 - boolToZ7 (rule110Step L true R) := by
  native_decide

-- NAND functional completeness (one named axiom; Sheffer 1913)
axiom boolean_nand_complete :
  forall (f : Bool -> Bool -> Bool),
    exists (circuit : Bool -> Bool -> Bool),
      forall (a b : Bool), circuit a b = f a b

-- Finite two-input NAND functional completeness (zero sorry given boolean_nand_complete)
-- States finite Boolean completeness only, not unbounded-tape Turing
-- universality; see the finite-completeness remark in the main text
-- (Section "NAND Functional Completeness and Turing Universality").
theorem rule110_center1_nand_functional_completeness :
  forall (f : Bool -> Bool -> Bool),
    exists (circuit : Bool -> Bool -> Bool),
      forall (a b : Bool), circuit a b = f a b :=
  boolean_nand_complete
```

Rule 110's Turing universality rests on Cook's 2004 construction; the conditional operational readback certificate for this is Rule110.CookUniversalityTop.cook_operational_stage3_tm_microstep_readback in rule110-lean (zero sorry modulo five named classical Cook collision-analysis bridge axioms), not the theorem listed above. This readback certificate is itself conditional on already-supplied compilations and does not by itself establish Turing universality; the general

TM $\rightarrow$ CTS compiler remains open. (The theorem above was formerly named `rule110_turing_universal_algebraic`; that name is retained in `rule110-lean` only as a deprecated alias, since it was misleading about the theorem's scope.)